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United States Patent	12387975
Kind Code	B2
Date of Patent	August 12, 2025
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Thermal process chamber lid with backside pumping

Abstract

Process chamber lid assemblies and process chambers comprising same are described. The lid assembly has a housing with a gas dispersion channel in fluid communication with a lid plate. A contoured bottom surface of the lid plate defines a gap to a top surface of a gas distribution plate. A pumping channel is formed between an upper outer peripheral contour of the gas distribution plate and the lid plate.

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Appl. No.: 18/208409

Filed: June 12, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230335434 A1	Oct. 19, 2023

Related U.S. Application Data

continuation parent-doc US 17720836 20220414 US 11715667 child-doc US 18208409
continuation parent-doc US 16886116 20200528 US 11335591 20220517 child-doc US 17720836
us-provisional-application US 62853699 20190528

Publication Classification

Int. Cl.: H01L21/768 (20060101); C23C16/02 (20060101); C23C16/455 (20060101); C23C16/458 (20060101); H01L21/02 (20060101); H01L21/285 (20060101); H01L21/48 (20060101); H01L21/687 (20060101); H01L23/532 (20060101)

U.S. Cl.:

CPC H01L21/76826 (20130101); C23C16/02 (20130101); C23C16/45553 (20130101); C23C16/4584 (20130101); H01L21/0228 (20130101); H01L21/28568 (20130101); H01L21/4871 (20130101); H01L21/68764 (20130101); H01L21/68771 (20130101); H01L21/76843 (20130101); H01L21/76861 (20130101); H01L21/76877 (20130101); H01L23/53266 (20130101);

Field of Classification Search

CPC: C23C (16/4412)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6106625	12/1999	Koai	118/724	C23C 16/455
9175394	12/2014	Yudovsky et al.	N/A	N/A
11335591	12/2021	Cui	N/A	C23C 16/4408
11715667	12/2022	Cui	438/656	H01L 21/68771
2002/0023907	12/2001	Morishige	219/121.84	G03F 1/72
2005/0208217	12/2004	Shinriki et al.	N/A	N/A
2007/0169373	12/2006	Aoki	34/443	F26B 3/20
2008/0202416	12/2007	Provencher et al.	N/A	N/A
2010/0166957	12/2009	Sneh	N/A	N/A
2013/0267045	12/2012	Lee et al.	N/A	N/A
2014/0103145	12/2013	White	239/548	C23C 16/45565
2015/0093913	12/2014	Toyoda et al.	N/A	N/A
2015/0340226	12/2014	Ashihara et al.	N/A	N/A
2016/0097119	12/2015	Cui et al.	N/A	N/A
2016/0312360	12/2015	Rasheed et al.	N/A	N/A
2016/0319424	12/2015	Takahashi	N/A	N/A
2017/0009347	12/2016	Jang et al.	N/A	N/A
2017/0101710	12/2016	Chandrasekharan	N/A	H01L 21/6719
2017/0283945	12/2016	Yahata et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2007270352	12/2006	JP	N/A
2010212335	12/2009	JP	N/A
2016204729	12/2015	JP	N/A
2017183383	12/2016	JP	N/A
100754537	12/2006	KR	N/A

OTHER PUBLICATIONS

Machine Translation of JP2010212335, 23 pages. cited by applicant

PCT International Search Report and Written Opinion in PCT/US2020/034903 dated Sep. 11, 2020, 13 pages. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/720,836, filed Apr. 14, 2022, which is a continuation of U.S. patent application Ser. No. 16/886,116, filed May 28, 2020, now U.S. Pat. No. 11,335,591, issued May 17, 2022, which claims priority to U.S. Provisional Application No. 62/853,699, filed May 28, 2019, the entire disclosures of which are hereby incorporated by reference herein.

TECHNICAL FIELD

(1) Embodiments of the present disclosure pertain to the field of electronic device manufacturing. In particular, embodiments of the disclosure are directed to apparatus for delivering reactive gases in semiconductor device manufacturing.

BACKGROUND

(2) Reliably producing submicron and smaller features is one of the key technologies for the next generation of very large scale integration (VLSI) and ultra large scale integration (ULSI) of semiconductor devices. However, as the fringes of circuit technology are pressed, the shrinking dimensions of interconnects in VLSI and ULSI technology have placed additional demands on the processing capabilities. The multilevel interconnects that lie at the heart of VLSI and ULSI technology use precise processing of high aspect ratio features, such as vias and other interconnects. Reliable formation of these interconnects is very important to VLSI and ULSI success and to the continued effort to increase circuit density and quality of individual substrates.

(3) As circuit densities increase, the widths of interconnects, such as vias, trenches, contacts, and other features, as well as the dielectric materials between, decrease while the thickness of the dielectric layers remain substantially constant, resulting in increased height-to-width aspect ratios of the features. Many traditional deposition processes have difficulty filling submicron structures and providing good step coverage for surface features.

(4) Atomic layer deposition (ALD) is a deposition technique being explored for the deposition of material layers over features having high aspect ratios. One example of an ALD process includes the sequential introduction of pulses of gases. For instance, one cycle for the sequential introduction of pulses of gases may contain a pulse of a first reactant gas, followed by a pulse of a purge gas and/or a pump evacuation, followed by a pulse of a second reactant gas, and followed by a pulse of a purge gas and/or a pump evacuation. The term “gas” as used herein is defined to include a single gas or a plurality of gases. Sequential introduction of separate pulses of the first reactant and the second reactant may result in the alternating self-limiting adsorption of monolayers of the reactants on the surface of the substrate and, thus, forms a monolayer of material for each cycle. The cycle may be repeated to form a film with a predetermined thickness. A pulse of a purge gas and/or a pump evacuation between the pulses of the first reactant gas and the pulses of the second reactant gas serves to reduce the likelihood of gas phase reactions of the reactants due to

excess amounts of the reactants remaining in the chamber.

(5) In some chamber designs for ALD processing, precursors and gases are delivered using a funnel lid through which precursor is distributed through multiple injectors above a funnel shaped lid. The injectors generate a circular motion of the injected gas which distributes through the funnel profile at the center of the lid. The rotational inertia of the gas/ALD precursor molecules distributes the molecules from center to edge resulting in improved uniformity deposition.

(6) It has been observed that reactive gases become trapped between the lid plate and the showerhead during processing resulting in non-uniformity of the deposited films. Accordingly, there is an ongoing need in the art for methods and apparatus to improve uniformity of deposited films.

SUMMARY

(7) One or more embodiments of the disclosure are directed to process chamber lid assemblies. A housing encloses a gas dispersion channel that extends along a central axis of the housing. The gas dispersion channel has an upper portion and a lower portion. A lid plate is coupled to the housing and has a contoured bottom surface that extends downwardly and outwardly from a central opening coupled to the lower portion of the gas dispersion channel to a peripheral portion of the lid plate. A gas distribution plate is disposed below the lid plate and has an upper outer peripheral contour configured to form a pumping channel between the gas distribution plate and the lid plate. The gas distribution plate has a top surface and a bottom surface with a plurality of apertures disposed through the gas distribution plate from the top surface to the bottom surface. The contoured bottom surface of the lid plate and top surface of the gas distribution plate define a gap.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) So that the manner in which the above recited features of the present disclosure can be understood in detail, a more particular description of the disclosure, briefly summarized above, may be had by reference to embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only typical embodiments of this disclosure and are therefore not to be considered limiting of its scope, for the disclosure may admit to other equally effective embodiments. The embodiments as described herein are illustrated by way of example and not limitation in the figures of the accompanying drawings in which like references indicate similar elements.

(2) FIG. 1 depicts a schematic view of a process chamber in accordance with some embodiments of the present disclosure;

(3) FIG. 2 depicts a schematic cross-sectional view of a process chamber in accordance with some embodiments of the present disclosure;

(4) FIG. 3 depicts a schematic cross-sectional view of a lid assembly in accordance with some embodiments of the present disclosure;

(5) FIGS. 4A-C depict schematic views of apertures disposed through a gas distribution plate in accordance with embodiments of the present disclosure;

(6) FIG. 5 shows a schematic cross-sectional view of a lid assembly in accordance with one or more embodiments of the disclosure;

(7) FIG. 6A shows a schematic cross-sectional view of a lid plate in accordance with one or more embodiments of the disclosure;

(8) FIG. 6B shows a schematic cross-sectional view of a lid assembly in accordance with one or more embodiments of the disclosure;

(9) FIG. 7 shows a schematic cross-sectional view of a lid assembly in accordance with one or more embodiment of the disclosure; and

(10) FIG. 8 shows a schematic cross-sectional view of a lid assembly in accordance with one or more embodiment of the disclosure.

(11) To facilitate understanding, identical reference numerals have been used, where possible, to designate identical elements that are common to the figures. The figures are not drawn to scale and may be simplified for clarity. Elements and features of one embodiment may be beneficially incorporated in other embodiments without further recitation.

DETAILED DESCRIPTION

(12) Before describing several exemplary embodiments of the disclosure, it is to be understood that the disclosure is not limited to the details of construction or process steps set forth in the following description. The disclosure is capable of other embodiments and of being practiced or being carried out in various ways.

(13) A “substrate” as used herein, refers to any substrate or material surface formed on a substrate upon which film processing is performed during a fabrication process. For example, a substrate surface on which processing can be performed include materials such as silicon, silicon oxide, strained silicon, silicon on insulator (SOI), carbon doped silicon oxides, amorphous silicon, doped silicon, germanium, gallium arsenide, glass, sapphire, and any other materials such as metals, metal nitrides, metal alloys, and other conductive materials, depending on the application. Substrates include, without limitation, semiconductor wafers. Substrates may be exposed to a pretreatment process to polish, etch, reduce, oxidize, hydroxylate, anneal and/or bake the substrate surface. In addition to film processing directly on the surface of the substrate itself, in the present disclosure, any of the film processing steps disclosed may also be performed on an under-layer formed on the substrate as disclosed in more detail below, and the term “substrate surface” is intended to include such under-layer as the context indicates. Thus for example, where a film/layer or partial film/layer has been deposited onto a substrate surface, the exposed surface of the newly deposited film/layer becomes the substrate surface.

(14) As used in this specification and the appended claims, the terms “precursor”, “reactant”, “reactive gas” and the like are used interchangeably to refer to any gaseous species that can react with the substrate surface.

(15) Embodiments of the present disclosure provide apparatus and methods that may be used to clean substrate processing chambers, such as an atomic layer deposition (ALD) chamber, and to deposit materials during, for example, an ALD process. Embodiments include substrate processing chambers and gas delivery systems which may include a remote plasma source and a gas distribution plate. The following process chamber description is provided for context and exemplary purposes, and should not be interpreted or construed as limiting the scope of the disclosure.

(16) FIG. 1 is a schematic view of a substrate processing chamber (process chamber **100**) including a gas delivery system **130** adapted for ALD processes in accordance with some embodiments of the present disclosure. FIG. 2 is a cross-sectional view of the process chamber **100**. Process chamber **100** includes a chamber body **102** having a processing volume within the chamber body **102** and beneath the chamber lid assembly **132**. Slit valve **108** in the process chamber **100** provides access for a robot (not shown) to deliver and retrieve a substrate **110**, such as a 200 mm or 300 mm semiconductor wafer or a glass substrate, to and from the process chamber **100**. A chamber liner **177** is disposed along the walls of the process chamber **100** to protect the chamber from corrosive gases used during processing/cleaning.

(17) A substrate support **112** supports the substrate **110** on a substrate receiving surface **111** in the process chamber **100**. The substrate support **112** is mounted to a lift motor **114** for raising and lowering the substrate support **112** and the substrate **110** disposed on the substrate support. A lift plate **116** (shown in FIG. 2), connected to a lift motor **118**, is mounted in the process chamber **100** to raise and lower lift pins **120** movably disposed through the substrate support **112**. The lift pins **120** raise and lower the substrate **110** over the surface of the substrate support **112**. The substrate

support **112** may include a vacuum chuck (not shown), an electrostatic chuck (not shown), or a clamp ring (not shown) for securing the substrate **110** to the substrate support **112** during a deposition process.

(18) The temperature of the substrate support **112** may be adjusted to control the temperature of the substrate **110**. For example, substrate support **112** may be heated using an embedded heating element, such as a resistive heater (not shown), or may be heated using radiant heat, such as heating lamps (not shown) disposed above the substrate support **112**. A purge ring **122** may be disposed on the substrate support **112** to define a purge channel **124** which provides a purge gas to a peripheral portion of the substrate **110** to prevent deposition on the peripheral portion of the substrate **110**.

(19) Gas delivery system **130** is disposed at an upper portion of the chamber body **102** to provide a gas, such as a process gas and/or a purge gas, to process chamber **100**. A vacuum system (not shown) is in communication with a pumping channel **179** to evacuate any desired gases from the process chamber **100** and to help maintain a desired pressure or pressure range inside the process chamber **100**.

(20) In some embodiments, the chamber lid assembly **132** includes a gas dispersion channel **134** extending through a central portion of the chamber lid assembly **132**. As shown in FIGS. **1** and **2**, the gas dispersion channel **134** extends perpendicularly toward the substrate receiving surface **111** and also extends along a central axis **133** of the gas dispersion channel **134**, through lid plate **170**, and to bottom surface **160**. In some embodiments, an upper portion of the gas dispersion channel **134** is substantially cylindrical along central axis **133** and a lower portion of the gas dispersion channel **134** tapers away from central axis **133**. The bottom surface **160** is sized and shaped to substantially cover the substrate **110** disposed on the substrate receiving surface **111** of the substrate support **112**. The bottom surface **160** tapers from an outer edge of the lid plate **170** towards the gas dispersion channel **134**. The gas delivery system **130** may provide one or more gasses to the gas dispersion channel **134** to process the substrate **110**. In some embodiments, the gas delivery system **130** may be coupled to the gas dispersion channel **134** via one gas inlet. In some embodiments, such as that shown in FIG. **3**, the gas delivery system may be coupled to the gas dispersion channel **134** via a plurality of inlets.

(21) As illustrated in FIG. **3**, circular gas flow **174**, which illustrates the flow of process gases through the gas dispersion channel **134**, may contain various types of flow patterns. In some embodiments, processing gases may be forced to make revolutions around central axis **133** of gas dispersion channel **134** while passing through the dispersion channel. In such embodiments, the circular gas flow **174** may contain various types of circular flow patterns, such as a vortex pattern, a helix pattern, a spiral pattern, or derivatives thereof.

(22) Although providing a circular gas flow **174** is beneficial for many applications, the inventors have discovered that in some applications, the circular gas flow can lead to non-uniform processing results. The inventors have observed the gas flow may lead to a donut-shaped deposition profile near a center of the substrate **110** being processed. The donut-shaped profile may be caused by the funnel shape of gas dispersion channel **134**. Therefore, in some embodiments, the process chamber **100** further includes a gas distribution plate **125** having a plurality of apertures **126** disposed through the gas distribution plate **125**. The gas distribution plate **125** extends to the surface of the gas dispersion channel **134** such that the only pathway from the gas dispersion channel **134** to the substrate is through the plurality of apertures **126** of the gas distribution plate **125**. The gas distribution plate **125** advantageously creates a choked flow of gas through the gas distribution plate **125** resulting in a more uniform deposition on the substrate **110** and, thus, substantially eliminating the donut-shaped deposition caused by the rotational flow of gas.

(23) In some embodiments, the gas distribution plate **125** is formed of a non-corrosive ceramic material such as, for example, aluminum oxide or aluminum nitride. In some embodiments, each of the plurality of apertures **126** may have an equivalent fluid conductance. In some embodiments, a density of the plurality of apertures **126** (e.g., the number of apertures or the size of the openings of

the apertures per unit area) may vary across the gas distribution plate **125** to achieve a desired deposition profile on the substrate **110**. For example, a higher density of apertures **126** may be disposed at a center of the gas distribution plate **125** to increase the deposition rate at the center of the substrate relative to the edge of the substrate to further improve deposition uniformity.

(24) Although the plurality of apertures **126** are depicted as cylindrical through holes, the plurality of apertures **126** may have different profiles. FIGS. **4A-C** depict different non-limiting embodiments of profiles of the plurality of apertures **126**. In the embodiment depicted in FIG. **4A**, the aperture **126** is a cylindrical through hole having curved edges **402** that surround the aperture. In the embodiment depicted in FIG. **4B**, the aperture **126** is a through hole having an upper portion **404** that tapers inwardly toward a center of the aperture, a cylindrical center portion **405** extending perpendicularly to an upper surface **127** of the gas distribution plate **125**, and a lower portion **406** that tapers outwardly from the center of the aperture. In the embodiment depicted in FIG. **4C**, the aperture **126** is a through hole having an upper portion **408** having a countersunk hole, a cylindrical center portion **409** extending perpendicularly to the upper surface **127** of the gas distribution plate **125**, and a lower portion **410** that tapers outwardly from the center of the aperture. Other profiles of the plurality of apertures **126** may alternatively be used to achieve optimal deposition uniformity during processing of the substrate **110**.

(25) Not wishing to be bound by theory, the inventors believe that the diameter of gas dispersion channel **134**, which is constant from the upper portion of gas dispersion channel **134** to a first point along central axis **133** and increasing from the first point to lower portion **135** of gas dispersion channel **134**, allows less of an adiabatic expansion of a gas through gas dispersion channel **134** which helps to control the temperature of the process gas contained in the circular gas flow **174**. For example, a sudden adiabatic expansion of a gas delivered into gas dispersion channel **134** may result in a drop in the temperature of the gas which may cause condensation of the gas and formation of droplets. On the other hand, a gas dispersion channel **134** that gradually tapers is believed to provide less of an adiabatic expansion of a gas. Therefore, more heat may be transferred to or from the gas, and, thus, the temperature of the gas may be more easily controlled by controlling the temperature of chamber lid assembly **132**. Gas dispersion channel **134** may gradually taper and contain one or more tapered inner surfaces, such as a tapered straight surface, a concave surface, a convex surface, or combinations thereof or may contain sections of one or more tapered inner surfaces (i.e., a portion tapered and a portion non-tapered).

(26) As shown in FIG. **3**, the upper portion of the gas dispersion channel **134** is defined by an insert **300** disposed in an inner region of a housing **375**. The insert **300** includes a cap **302** at an upper portion of the insert **300** and a central passageway that at least partially defines the gas dispersion channel **134**. The cap **302** extends over the housing **375** to hold the insert **300** in place. The insert **300** and cap **302** include a plurality of o-rings **385** disposed between the insert **300** and the housing **375** to ensure proper sealing. The insert **300** includes a plurality of circumferential apertures which, when the insert **300** is inserted into the housing **375**, form a corresponding plurality of circumferential channels **360**, **365**, **370**. The plurality of circumferential channels **360**, **365**, **370** are fluidly coupled to the gas dispersion channel **134** via a corresponding plurality of holes **340**, **345**, **350**. In the embodiment shown in FIG. **3**, the gas delivery system **130** is coupled to the gas dispersion channel **134** via a plurality of gas feed lines **310**, **315**, **320**. The gas feed lines **310**, **315**, **320** are fluidly coupled to the plurality of circumferential channels **360**, **365**, **370** to provide one or more gases to the gas dispersion channel **134**.

(27) Returning to FIGS. **1** and **2**, the process chamber **100** further includes a chamber cleaning system including a remote plasma source (RPS) **190**, an isolation collar **192** coupled to the RPS **190** at one end and the cap **302** at an opposite end, a heater plate **198** coupled to an upper surface of the lid plate **170**, and a cleaning gas (i.e., purge gas) source **197** fluidly coupled to the RPS **190**. The cleaning gas source may include any gas suitable for forming a plasma to clean the process chamber **100**. In some embodiments, for example, the cleaning gas may be nitrogen trifluoride

(NF.sub.3). The isolation collar **192** includes an inner channel **193** that is fluidly coupled to the gas dispersion channel **134** through a plurality of holes **285** disposed in a central portion of the cap **302** to flow a plasma from the RPS **190** through the gas dispersion channel **134** and into the reaction zone **164**. The heater plate **198** may be formed of stainless steel and include a plurality of resistive heating elements dispersed throughout the plate.

(28) Typically, a cleaning gas is flowed through the gas dispersion channel **134** and the reaction zone **164** after a first gas is provided to the gas dispersion channel **134** by the gas delivery system **130** to quickly purge the first gas from the gas dispersion channel **134** and the reaction zone **164**. Subsequently, a second gas is provided by the gas delivery system **130** to the gas dispersion channel **134** and the cleaning gas is again flowed through the gas dispersion channel **134** to the reaction zone **164** to quickly purge the second gas from the gas dispersion channel **134** and the reaction zone **164**. However, the addition of the gas distribution plate **125** chokes the flow of the cleaning gas to the pumping channel **179** and prolongs the cleaning process. As such, the inventors have incorporated an exhaust system **180** having an exhaust conduit **184** coupled to the isolation collar **192** at a first end **186** and to the pumping channel **179** at a second end **188**. A valve **182** is disposed in the exhaust conduit **184** to selectively fluidly couple the exhaust conduit **184** to the inner channel **193**. In some embodiments, for example, the valve **182** may be a plunger type valve having a plunger **202** that is moveable between a first position (shown in FIG. 2) to fluidly couple the exhaust conduit **184** to the inner channel **193** and a second position to seal off the exhaust conduit **184** from the inner channel **193**. Each time the cleaning gas is flowed through the gas dispersion channel **134** and the reaction zone **164**, the valve **182** is opened and the cleaning gas is rapidly exhausted to the pumping channel **179**.

(29) When a pressure inside of the process chamber **100** exceeds a pressure inside of the RPS **190**, processing gasses may flow up to and damage the RPS **190**. The plurality of holes **285** serve as a choke point to prevent a backflow of processing gases from flowing up through the inner channel **193** and into the RPS **190**. The isolation collar **192** may be formed of any material that is non-reactive with the cleaning gas being used. In some embodiments, the isolation collar **192** may be formed of aluminum when the cleaning gas is NF.sub.3. In some embodiments, the isolation collar **192** and the insert **300** may be formed of aluminum and coated with a coating to prevent corrosion of the isolation collar **192** and the insert **300** from corrosive gases when used. For example, the coating may be formed of nickel or aluminum oxide.

(30) Referring to FIG. 3, the RPS **190** operates at a temperature less than or equal to about 40° C. In order advantageously insulate the RPS **190** from heat generated in the process chamber **100**, a thermal isolation ring **394** is disposed between the isolation collar **192** and the cap **302**. The thermal isolation ring **394** is formed of a metal with low thermal conductivity (e.g., lower than the thermal conductivity of the isolation collar **192** and the cap **302**). In addition, an o-ring **385** may also be disposed between the isolation collar **192** and the cap **302** to further reduce the contact area between the isolation collar **192** and the cap **302**. The combination of the thermal isolation ring **394** and the o-ring **385** acts as a thermal choke to ensure that the heat generated in the process chamber **100** does not adversely affect the RPS **190**.

(31) In some embodiments, when the lid plate **170** is heated above 100° C. the process chamber **100** may include a differential pumping line **250** to ensure that any process gases or byproducts trapped between o-rings **385** are exhausted to the pumping channel **179**. The differential pumping line **250** is coupled to the lid plate **170** at a first end and to the housing **375** at a second end opposite the first end. The differential pumping line is fluidly coupled to the gas dispersion channel **134** and to one or more channels **260** formed at areas between two or more o-rings **385**. When the valve **182** is opened to exhaust the gas dispersion channel **134**, the differential pumping line exhausts gases trapped between o-rings **385**.

(32) Returning to FIG. 3, a portion of bottom surface **160** of chamber lid assembly **132** may be contoured or angled downwardly and outwardly from a central opening coupled to the gas

dispersion channel **134** to a peripheral portion of chamber lid assembly **132** to help provide an improved velocity profile of a gas flow from gas dispersion channel **134** across the surface of substrate **110** (i.e., from the center of the substrate to the edge of the substrate). Bottom surface **160** may contain one or more surfaces, such as a straight surface, a concave surface, a convex surface, or combinations thereof. In one embodiment, bottom surface **160** is convexly funnel-shaped.

(33) In one example, bottom surface **160** is downwardly and outwardly sloping toward an edge of the substrate receiving surface **111** to help reduce the variation in the velocity of the process gases traveling between bottom surface **160** of chamber lid assembly **132** and substrate **110** while assisting to provide uniform exposure of the surface of substrate **110** to a reactant gas. The components and parts of chamber lid assembly **132** may contain materials such as stainless steel, aluminum, nickel-plated aluminum, nickel, alloys thereof, or other suitable materials. In one embodiment, lid plate **170** may be independently fabricated, machined, forged, or otherwise made from a metal, such as aluminum, an aluminum alloy, steel, stainless steel, alloys thereof, or combinations thereof.

(34) In some embodiments, inner surface **131** of gas dispersion channel **134** and bottom surface **160** of chamber lid assembly **132** may contain a mirror polished surface to help a flow of a gas along gas dispersion channel **134** and bottom surface **160** of chamber lid assembly **132**.

(35) Referring to FIGS. 1-3, in a processing operation, substrate **110** is delivered to process chamber **100** through slit valve **108** by a robot (not shown). Substrate **110** is positioned on substrate support **112** through cooperation of lift pins **120** and the robot. Substrate support **112** raises substrate **110** into close opposition to a lower surface of the gas distribution plate **125**. A first gas flow may be injected into gas dispersion channel **134** of process chamber **100** by the gas delivery system **130** together or separately (i.e., pulses) with a second gas flow. The first gas flow may contain a continuous flow of a purge gas from a purge gas source and pulses of a reactant gas from a reactant gas source or may contain pulses of a reactant gas from the reactant gas source and pulses of a purge gas from the purge gas source. The second gas flow may contain a continuous flow of a purge gas from a purge gas source and pulses of a reactant gas from a reactant gas source or may contain pulses of a reactant gas from a reactant gas source and pulses of a purge gas from a purge gas source.

(36) The circular gas flow **174** travels through gas dispersion channel **134** and subsequently through the plurality of apertures **126** in the gas distribution plate **125**. The gas is then deposited on the surface of substrate **110**. Bottom surface **160** of chamber lid assembly **132**, which is downwardly sloping, helps reduce the variation of the velocity of the gas flow across the surface of gas distribution plate **125**. Excess gas, byproducts, etc. flow into the pumping channel **179** and are then exhausted from process chamber **100**. Throughout the processing operation, the heater plate **198** may heat the chamber lid assembly **132** to a predetermined temperature to heat any solid byproducts that have accumulated on walls of the process chamber **100** (or a processing kit disposed in the chamber). As a result, any accumulated solid byproducts are vaporized. The vaporized byproducts are evacuated by a vacuum system (not shown) and pumping channel **179**. In some embodiments, the predetermined temperature is greater than or equal to 150° C.

(37) Some process conditions can cause step coverage issues due to, for example, residual precursors in the gas delivery system allowing gas phase reactions. In a typical ALD process, gas phase reactions are generally avoided. Accordingly, some embodiments of the disclosure provide process chamber lids and processing chambers with backside pumping capability to a chamber lid. The apparatus of some embodiments is a thermal chamber lid with no plasma source connected thereto. In some embodiments, the chamber lid is configured with a remote plasma source to provide a remote plasma to the process chamber.

(38) One or more embodiments of the disclosure advantageously provide apparatus to improve step coverage of films on surface features. One or more embodiments of the disclosure advantageously provide apparatus that adds backside pumping to remove residual reactive gases. In some

embodiments, the apparatus helps pump chemicals trapped between the lid plate and the showerhead more efficiently.

(39) FIG. 5 shows a process chamber lid assembly **500** according to one or more embodiment of the disclosure. A housing **375** encloses a gas dispersion channel **134** that extends along a central axis **133** of the housing **375**. The gas dispersion channel **134** has an upper portion **134a** and a lower portion **134b**.

(40) A lid plate **170** is coupled to the housing **375** and has a contoured bottom surface **160**. The contoured bottom surface **160** extends downwardly and outwardly from a central opening **136** coupled to the lower portion **134b** of the gas dispersion channel **134** to an outer peripheral portion **138** of the lid plate **170**. In the illustrated embodiment, the outer peripheral portion **138** refers to the outer portion of the contoured bottom surface **160** adjacent the outer peripheral edge **137**.

(41) The lid assembly **500** includes a gas distribution plate **125** disposed below the lid plate **170**. The gas distribution plate **125** has a top surface **128** and a bottom surface **129** with a plurality of apertures **126** disposed through the gas distribution plate **125** from the top surface **128** to the bottom surface **129**.

(42) The gas distribution plate **125** has an upper outer peripheral contour **520** configured to form a pumping channel **530** between the gas distribution plate **125** and the lid plate **170**. The pumping channel **530** shown in the embodiment of FIG. 5 is defined between an outer peripheral bottom surface **532** of the lid plate **170** and the upper outer peripheral contour **520** of the gas distribution plate **125**. In some embodiments, the outer peripheral bottom surface **532** of the lid plate **170** is further from the central axis **133** than the outer peripheral portion **138** of the contoured bottom surface **160**. Stated differently, in some embodiments, the outer peripheral bottom surface **532** surrounds the contoured bottom surface **160**.

(43) The contoured bottom surface **160** of the lid plate **170** and the top surface **128** of the gas distribution plate **125** define a gap G. As the bottom surface **160** is contoured, the gap G is variable as a function of distance from the central axis **133**. In some embodiments, the inner zone Z.sub.I has a larger gap than the middle zone Z.sub.M, and the middle zone Z.sub.M has a larger gap than the outer zone Z.sub.O. FIG. 6A shows a schematic cross-sectional view of the lid plate **170** with a contoured bottom surface **160** similar to that shown in FIG. 5. FIG. 6B shows a schematic cross-sectional view of the lid plate **170** and gas distribution plate **125** of FIG. 5 showing the relationship of the gap G to the radial distance from the central axis **133**. In FIG. 6A, the contoured bottom surface **160** of the lid plate **170** is separated into three zones: an inner zone Z.sub.I; a middle zone Z.sub.M; and an outer zone Z.sub.O. In this embodiment, in the middle zone Z.sub.M the contoured bottom surface **160** is flat so that the gap GM is uniform. Referring to FIG. 6B, which shows a partial cross-sectional view, the gap GM in the middle zone Z.sub.M is uniform from the left edge to the right edge of the middle zone Z.sub.M. In the inner zone Z.sub.I, the gap G.sub.I is a function of the distance x measured from the central axis **133**. FIG. 6B shows two measurements for the gap G.sub.Idx in the inner zone Z.sub.I. In the outer zone Z.sub.O, the gap G.sub.O is a function of the distance x measured from the central axis **133**. One measurement is shown in FIG. 6B for the gap G.sub.Odx in the outer zone Z.sub.O. The skilled artisan will recognize that the illustrated measurements are for descriptive purposes only. The apertures **126** in the gas distribution plate **125** in FIG. 6B are omitted for descriptive purposes.

(44) Referring again to FIG. 6B, in some embodiments, the inner zone Z.sub.I is defined from a central axis **133** of the lid plate **170** to an inner zone radial distance RI from the central axis **133**. The middle zone Z.sub.M is defined from the inner zone radial distance R.sub.I to a middle zone radial distance R.sub.M from the central axis **133**. The outer zone Z.sub.O is measured from the middle zone radial distance Z.sub.M to an outer zone radial distance R.sub.O at an outer peripheral edge **137** of the contoured bottom surface **160**.

(45) The size of the middle zone Z.sub.M can be any suitable size measured relative to the total radial distance from the central axis **133** to the outer peripheral edge **137**. In some embodiments,

the distance from the central axis **133** to the outer peripheral edge **137** is greater than or equal to about 50 mm, 100 mm, 150 mm or 200 mm. In some embodiments, the distance from the central axis **133** to the outer peripheral edge **137** is greater than the radius of a substrate to be processed. For example, in an embodiment where a 300 mm substrate is being processed, the radial distance from the central axis to the edge of the substrate is, assuming the substrate is centered, 150 mm. In this example, the distance from the central axis **133** to the outer peripheral edge **137** is greater than or equal to 150 mm.

(46) In some embodiments, the distance from the central axis **133** to the middle zone radial distance Z.sub.M is greater than or equal to about 50 mm, 100 mm, 150 mm or 200 mm. In some embodiments, the distance from the central axis **133** to the middle zone radial distance Z.sub.M is greater than the radius of a substrate to be processed. For example, in an embodiment where a 300 mm substrate is being processed, the radial distance from the central axis to the edge of the substrate is, assuming the substrate is centered, 150 mm. In this example, the distance from the central axis **133** to the middle zone radial distance Z.sub.M is greater than or equal to 150 mm, for example.

(47) In some embodiments, the size of the middle zone Z.sub.M of the lid plate **170** is in the range of about 10% to about 90% of the distance from the central axis to the outer zone radial distance R.sub.O. In some embodiments, the size of the middle zone Z.sub.M of the lid plate **170** is in the range of about 20% to about 80%, or in the range of about 30% to about 70%, or in the range of about 40% to about 60% of the distance from the central axis **133** to the outer zone radial distance R.sub.O.

(48) In some embodiments, substantially uniform gap in the middle zone Z.sub.M is in the range of about 0.1 inches to about 2 inches (about 2.5 mm to about 51 mm). As used in this manner, the term “substantially uniform gap” means that the gap at any radial distance within the middle zone Z.sub.M is within 5%, 2%, 1% or 0.5% of the average gap in the middle zone Z.sub.M.

(49) In some embodiments, the outer zone Z.sub.O is sloped from the middle zone Z.sub.M to a front face **161** of the lid plate **170**. In some embodiments, the outer zone Z.sub.O is sloped from the middle zone Z.sub.M to the top surface **128** of the gas distribution plate **125**. The slope of the outer zone Z.sub.O in relation to the flat middle zone Z.sub.M forms an outer zone angle Θ , as shown in FIG. 7. In some embodiments, the outer zone supplementary angle is in the range of about 15° to about 75°, or in the range of about 30° to about 60°, or in the range of about 40° to about 50°. In some embodiments, the outer zone angle is in the range of about 105° to about 165°, or in the range of about 120° to about 150°, or in the range of about 130° to about 140°.

(50) Referring to FIGS. 5 and 7, in some embodiments, the outer zone Z.sub.O of the contoured bottom surface **160** is connected to the pumping channel **530** through pumping holes **525** formed in the lid plate **170**. In some embodiments, the pumping holes **525** are formed in the outer zone Z.sub.O of the contoured bottom surface **160**. The number of pumping holes **525** can be varied based on, for example, the size of the lid plate **170**. In some embodiments, there are in the range of about 24 to about 144 pumping holes **525**.

(51) As shown in FIG. 7, the pumping holes intersect the outer zone Z.sub.O of the contoured bottom surface **160** at an angle ϕ . In some embodiments, the angle ϕ is in the range of about 75° to about 105°, or in the range of about 80° to about 100°, or in the range of about 85° to about 95°, or in the range of about 88° to about 92°.

(52) The lid assembly **500** of some embodiments includes at least one pump port **560** in fluid communication with the pumping channel **530**, as shown in FIG. 5. The pump port **560** can be a separate component connected to the lid plate **170**. In some embodiments, there are two or more pump ports **560** connected to the pumping channel **530** at different radial positions. In some embodiments, each of the pump ports is connected to a separate vacuum source for evacuation purposes. In some embodiments, the pump ports are in fluid communication with a single vacuum source. In some embodiments, the pump ports are in fluid communication with the pumping

channel **179** (see FIG. **1**).

(53) FIG. **8** shows a lid assembly **600** in accordance with one or more embodiment of the disclosure. The contoured bottom surface **160** of the lid plate **170** is sloped from an inner edge **610** of the contoured bottom surface **160** to the outer peripheral edge **137**.

(54) The slope of the contoured bottom surface **160** creates a gap **G** that decreases to a minimum at the outer peripheral edge **137**. In some embodiments, the minimum gap **G** is in the range of about 0.01 inches to about 1 inches (about 0.25 mm to about 25.4 mm), or in the range of about 0.05 inches to about 0.5 inches (about 1.25 mm to about 12.7 mm).

(55) Additional embodiments of the disclosure are directed to processing chambers incorporating lid assembly **500** or lid assembly **600**.

(56) In the foregoing specification, embodiments of the disclosure have been described with reference to specific exemplary embodiments thereof. It will be evident that various modifications may be made thereto without departing from the broader spirit and scope of the embodiments of the disclosure as set forth in the following claims. The specification and drawings are, accordingly, to be regarded in an illustrative sense rather than a restrictive sense.

Claims

1. A process chamber lid assembly comprising: a lid plate comprising a contoured bottom surface that extends downwardly and outwardly from a central opening around a central axis of the lid plate to a peripheral portion of the lid plate, the contoured bottom surface comprising an inner zone, a middle zone and an outer zone, the inner zone defined from the central axis to an inner zone radial distance from the central axis, the middle zone defined from the inner zone radial distance to a middle zone radial distance, the outer zone defined from the middle zone radial distance to an outer zone radial distance at an outer peripheral edge of the contoured bottom surface, the outer zone sloped from the middle zone radial distance to a front face of the lid plate forming an outer zone angle in the range of 105° to 165°, the outer zone comprising a plurality of pumping holes formed through the lid plate; a gas distribution plate disposed below the lid plate, the gas distribution plate having an upper outer peripheral contour configured to form a pumping channel between the gas distribution plate and the lid plate, the gas distribution plate having a top surface and a bottom surface with a plurality of apertures disposed through the gas distribution plate from the top surface to the bottom surface; and at least one pump port through the lid plate and in fluid communication with the pumping channel.

2. The process chamber lid assembly of claim 1, wherein when the gas distribution plate is disposed adjacent the front face of the lid plate, a gap between the inner zone of the contoured bottom surface and the gas distribution plate is larger than a gap between the middle zone of the contoured bottom surface and the gas distribution plate, which is larger than a gap between the outer zone of the contoured bottom surface and the gas distribution plate.

3. The process chamber lid assembly of claim 2, wherein the middle zone of the contoured bottom surface forms a substantially uniform gap.

4. The process chamber lid assembly of claim 3, wherein the substantially uniform gap is in the range of about 0.1 inch to about 2 inches.

5. The process chamber lid assembly of claim 2, wherein the middle zone comprises in the range of about 10% to about 90% of a distance from the central axis to the outer zone radial distance.

6. The process chamber lid assembly of claim 1, wherein the pumping holes intersect the outer zone of the contoured bottom surface at an angle in the range of about 85° to about 95°.

7. The process chamber lid assembly of claim 1, wherein there are in the range of about 24 to about 144 pumping holes in the outer zone.

8. A processing chamber comprising the process chamber lid assembly of claim 1.

9. The processing chamber of claim 8, further comprising a housing enclosing a gas dispersion

channel that extends along a central axis of the housing, the gas dispersion channel aligned with the central opening of the lid plate.

10. The processing chamber of claim 9, further comprising a remote plasma source fluidly coupled to the gas dispersion channel.

11. The processing chamber of claim 9, further comprising two pump ports connected to the pumping channel at different radial positions on the lid plate and a vacuum source configured to provide backside pumping to remove trapped chemicals between the lid plate and the gas distribution plate.

12. A processing chamber comprising: a lid plate comprising a contoured bottom surface that extends downwardly and outwardly from a central opening around a central axis of the lid plate to a peripheral portion of the lid plate, the contoured bottom surface comprising an inner zone, a middle zone and an outer zone, the inner zone defined from the central axis to an inner zone radial distance from the central axis, the middle zone defined from the inner zone radial distance to a middle zone radial distance, the outer zone defined from the middle zone radial distance to an outer zone radial distance at an outer peripheral edge of the contoured bottom surface, the outer zone sloped from the middle zone radial distance to a front face of the lid plate forming an outer zone angle in the range of 105° to 165° , the outer zone comprising a plurality of pumping holes formed through the lid plate; a gas distribution plate disposed below the lid plate, the gas distribution plate having an upper outer peripheral contour configured to form a pumping channel between the gas distribution plate and the lid plate, the gas distribution plate having a top surface and a bottom surface with a plurality of apertures disposed through the gas distribution plate from the top surface to the bottom surface; and a housing enclosing a gas dispersion channel that extends along a central axis of the housing, the gas dispersion channel aligned with the central opening of the lid plate.

13. The processing chamber of claim 12, wherein when the gas distribution plate is disposed adjacent the front face of the lid plate, a gap between the inner zone of the contoured bottom surface and the gas distribution plate is larger than a gap between the middle zone of the contoured bottom surface and the gas distribution plate, which is larger than a gap between the outer zone of the contoured bottom surface and the gas distribution plate.

14. The processing chamber of claim 13, wherein the middle zone of the contoured bottom surface forms a substantially uniform gap.

15. The processing chamber of claim 14, wherein the substantially uniform gap is in the range of about 0.1 inch to about 2 inches.

16. The processing chamber of claim 13, wherein the middle zone comprises in the range of about 10% to about 90% of a distance from the central axis to the outer zone radial distance.

17. The processing chamber of claim 12, wherein the pumping holes intersect the outer zone of the contoured bottom surface at an angle in the range of about 85° to about 95° .

18. The processing chamber of claim 12, wherein there are in the range of about 24 to about 144 pumping holes in the outer zone.

19. The processing chamber of claim 12, further comprising a remote plasma source fluidly coupled to the gas dispersion channel.
